

PROMPTMINER-SAM2: PROMPT MINING FOR REMOTE SENSING INSTANCE SEGMENTATION

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ABSTRACT

Remote-sensing scenes often contain visually similar or adjacent instances that are difficult to separate under shadows and narrow inter-instance gaps. Common proposal-based methods can localize instances effectively, but a proposal box alone provides insufficient shape cues for accurate instance-mask prediction. To address this issue, we propose PromptMiner-SAM2, a two-stage instance segmentation framework that mines proposal-level shape cues and injects them into prompt-conditioned SAM2 mask decoding. The coarse shape cues are first mined and serve as complementary instance-specific prompts for SAM2 mask prediction. To refine instance boundaries, an uncertainty-guided decoder-tail module is designed to identify uncertain local pixels and correct their predictions using local features and instance context. Experiments on the WHU and NWPU VHR-10 benchmarks show that PromptMiner-SAM2 achieves mask AP values of 73.4% and 68.3%, respectively, attaining the highest mask AP among the compared methods.

Index Terms— Instance segmentation, remote sensing imagery, SAM2, prompt mining, mask refinement

1. INTRODUCTION

Remote-sensing instance segmentation supports cadastral mapping, urban planning, and post-disaster assessment [1], [2]. In aerial imagery, shadows and low contrast often obscure object boundaries, while narrow inter-instance gaps leave little visual separation between adjacent instances. After localization, these ambiguities still need to be resolved during mask prediction, where a rectangular proposal box provides only a coarse spatial extent and insufficient shape cues for accurate instance-mask prediction. Figure 1 illustrates this limitation: the proposal box covers neighboring buildings with similar appearance and a narrow contact region, while the ground-truth masks distinguish them as separate instances.

Existing remote-sensing instance segmentation methods couple instance discovery with mask prediction through two-stage, one-stage, cascade, or conditional designs [3]–[6]. Transformer-style frameworks provide another modeling route by using global sequence representations for set-based prediction [7]–[9]. Alternative instance representations have

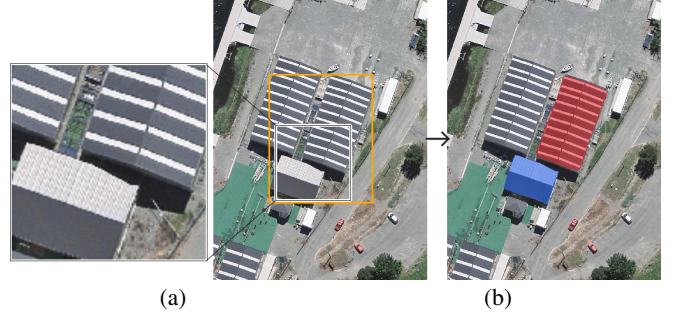


Fig. 1. An example of ambiguous instance separation in aerial imagery. (a) A proposal box covers neighboring buildings with similar appearance and a narrow contact region, as shown in the zoomed view. (b) Ground-truth masks distinguish the neighboring buildings as separate instances with clear boundaries.

also been explored through diffusion-based filter denoising and architecture search for efficient prediction [18]–[19]. These approaches generally rely on rectangular proposals or generic instance queries, which provide limited target-shape cues when neighboring objects have similar appearance. Beyond these architectures, prompt-conditioned foundation models offer a complementary way to inject spatial constraints into mask decoding. SAM and SAM2 accept sparse or dense prompts, motivating remote-sensing adaptation and automatic prompt learning such as RSPrompter [10]–[15]. Related refinement methods further exploit spatial cues: SAMRefiner derives multiple prompt types from an initial mask; PointRend focuses prediction on uncertain locations [16], [17]. These studies highlight the value of explicit spatial cues for resolving ambiguous instance boundaries, motivating their closer integration with proposal-based mask prediction.

These lines of work point to a common opportunity: prompt-conditioned decoders can benefit from instance-specific spatial cues beyond a rectangular extent. In a proposal-driven SAM2 pipeline, the detector already produces proposal-level features, but a box alone does not fully convey the shape of the localized instance for mask prediction. Preserving and transforming these cues into prompt-compatible guidance therefore becomes important

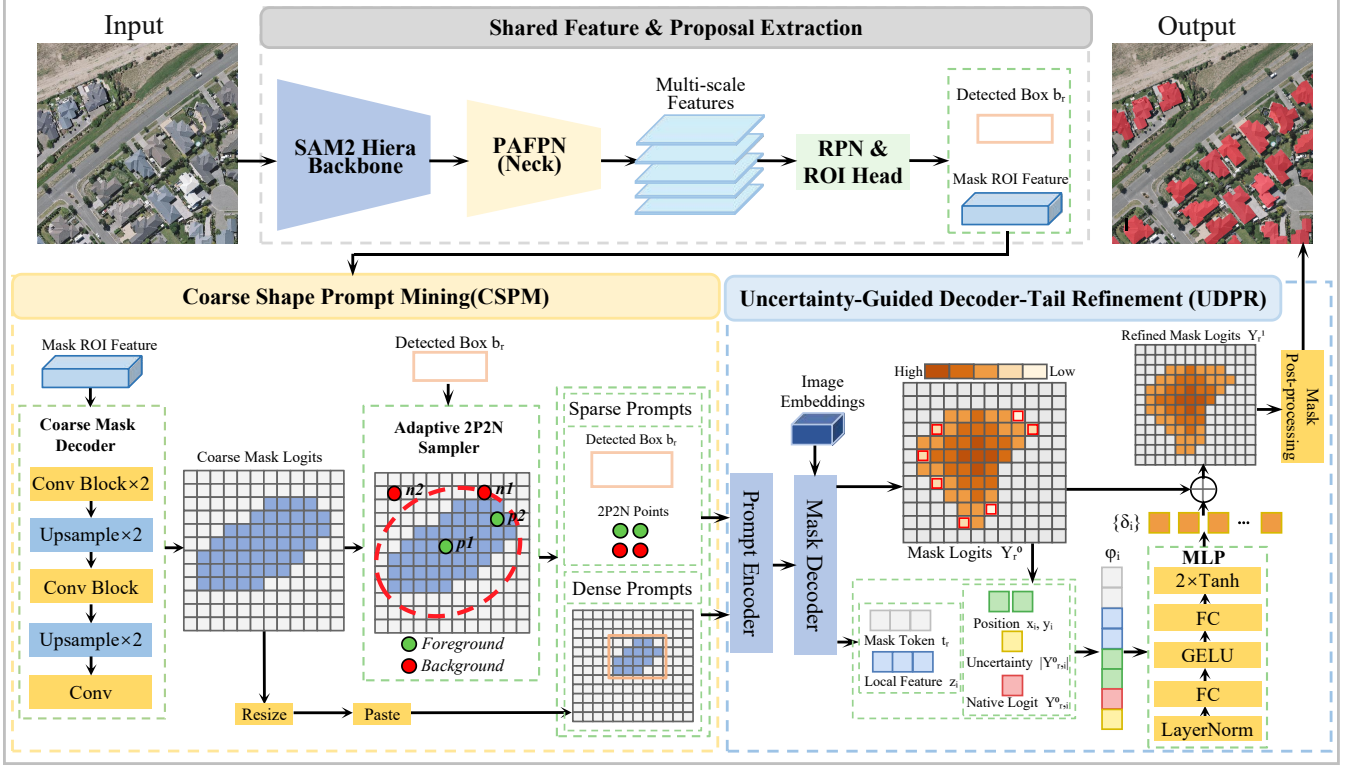


Fig. 2. Overview of the proposed PromptMiner-SAM2. SAM2 Hiera features support proposal extraction and provide image embeddings for mask decoding. Each detected box and its mask ROI feature drive coarse-shape prompt mining, which produces point, box, and dense-mask prompts. Uncertainty-guided decoder-tail refinement updates selected uncertain logits before mask post-processing.

for connecting instance discovery with accurate SAM2 mask decoding. To this end, we propose PromptMiner-SAM2, a two-stage framework that integrates proposal-based instance discovery with prompt-conditioned SAM2 segmentation.

Specifically, our contributions are threefold:

- We propose a proposal-driven two-stage framework for remote-sensing instance segmentation, coupling SAM2 with detector-based proposal extraction to connect instance discovery with prompt-conditioned mask decoding.
- We propose the coarse-shape prompt mining module (CSPM), which converts proposal-level shape estimates into complementary prompts to preserve instance-specific cues for separating adjacent objects.
- We design the uncertainty-guided decoder-tail refinement module (UDPR), which targets ambiguous mask locations with localized residual corrections to improve instance-mask delineation.

2. METHOD

As depicted in Fig. 2, PromptMiner-SAM2 couples a two-stage detector with the SAM2 prompt-to-mask interface. Hiera features feed the PAFPN/RPN/RoI pathway for proposal extraction, while the SAM2 mask decoder combines the image embedding with proposal-conditioned prompts. The resulting decoder features and mask logits are refined at uncertain locations, yielding the final instance masks for each detected proposal after post-processing.

2.1. Coarse-Shape Prompt Mining

CSPM converts each proposal’s ROI feature into a local shape estimate and uses it to construct complementary geometric cues for SAM2. It outputs 2P2N point prompts, the proposal box, and a dense shape prompt in the image-aligned coordinate frame.

For each proposal, the coarse branch maps its ROI feature X_r to a proposal-local shape logit map

$$C_r = f_{\text{coarse}}(X_r), \quad C_r \in \mathbb{R}^{1 \times 64 \times 64}. \quad (1)$$

Let $p_r = \sigma(C_r)$. PromptMiner derives a foreground support at probability 0.5 from p_r , while retaining the signed log-

its C_r for the dense prompt. It mines two positive and two negative points (2P2N) from this shared coarse shape. The first positive point is selected among confident foreground interior pixels ($p_r \geq 0.6$) by its foreground confidence weighted by its normalized distance to the support boundary; the second follows the same criterion while remaining spatially separated from the first. The first negative point comes from the one-pixel outer ring with low foreground probability ($p_r \leq 0.3$). The second is selected from background outside a dilation of the foreground support and is separated from the first positive and ring-negative points.

These four points encode complementary interior, exterior, and background cues. The 2P2N set maps them to signed prompts in the image frame.

Coordinates map to the image frame; signed logits are resized and pasted into b_r 's mask-input canvas:

$$S_r = \mathcal{M}(C_r, b_r), \quad Q_r = \text{Place}(C_r, b_r), \quad (2)$$

Here, $\mathcal{M}$ denotes 2P2N selection and coordinate mapping, and Place is the resize-and-paste operation. The two corners of b_r provide the box prompt. SAM2 jointly encodes the point, box, and dense-mask prompts, then decodes them with the image embedding in one pass:

$$Y_r^0 = \mathcal{D}(\mathcal{E}_I(I), \tilde{\mathcal{E}}_P(S_r, b_r, Q_r)), \quad (3)$$

where $\mathcal{E}_I$, $\tilde{\mathcal{E}}_P$, and $\mathcal{D}$ denote the SAM2 image encoder (including its decoder-side high-resolution features), complete SAM2 prompt-embedding pathway, and mask decoder, respectively. The pathway $\tilde{\mathcal{E}}_P$ includes the SAM2 composition of the dense-mask embedding. Q_r retains the signed rather than binarized coarse logits. Thus, the dense prompt preserves the spatial variation of the proposal-local shape, while the point and box prompts provide discrete geometric constraints to the same decoder.

2.2. Uncertainty-Guided Refinement

UDPR focuses on ambiguous locations in the initial SAM2 mask. It combines local decoder features with instance-level context to predict localized residual corrections. The same decoding pass returns an upscaled decoder feature map U_r , the first output mask token t_r , and decoder logit map Y_r^0 . On the raster grid $\mathcal{G}_r$, UDPR visits the K_r locations with the smallest logit magnitudes:

$$K_r = \min(K, |\mathcal{G}_r|), \quad (4)$$

$$\Omega_r = \text{BottomK}(|Y_r^0|, K_r), \quad K = 64,$$

where BottomK returns the K_r indices in $\mathcal{G}_r$ with the smallest logit magnitudes. For each $i \in \Omega_r$, the aligned feature z_i is gathered from U_r and combined with the token, normalized location, local logit, and logit magnitude:

$$\phi_i = [z_i, t_r, Y_{r,i}^0, x_i, y_i, |Y_{r,i}^0|]. \quad (5)$$

The local feature describes the decoder output near i , the mask token conditions the update on the decoded instance, and the coordinate and logit terms specify the position and current decision state. A lightweight MLP g then predicts a bounded residual for each selected location:

$$\delta_i = 2 \tanh g(\phi_i), \quad i \in \Omega_r, \quad (6)$$

The refined logit map is formed by scattering these residuals back to their original positions:

$$Y_{r,i}^1 = \begin{cases} Y_{r,i}^0 + \delta_i, & i \in \Omega_r, \\ Y_{r,i}^0, & i \notin \Omega_r. \end{cases} \quad (7)$$

This selective update leaves all nonselected decoder logits unchanged and applies bounded residual corrections on the existing SAM2 decoder grid.

3. EXPERIMENTS

3.1. Experimental Setup

Experiments use the WHU dataset [20], a single-class building benchmark with 2,943/627/2,220 training/validation/test images, and the NWPU VHR-10 dataset [21], a ten-class benchmark. Because the latter provides detection annotations only, we use the publicly released instance masks of Su *et al.* [22] and its 520/130 train/evaluation split.

Our PyTorch implementation uses distributed data parallelism on four NVIDIA RTX 3090 GPUs, with one image per GPU and two-step gradient accumulation (effective batch size eight). Both benchmark runs initialize from SAM2 Hiera-Base+ weights: base Hiera parameters remain frozen, while its LoRA adapters and the mask decoder are updated. We use AdamW with an initial learning rate of 5×10^{-4} and cosine annealing; training uses cross-entropy, Smooth L1, binary cross-entropy, and Dice losses. A value of $K = 64$ is selected by screening $K \in \{32, 64, 128\}$ under a common protocol; it achieves the highest overall bbox AP and mask AP in Table 1 and is used in the main experiments.

Table 1. Sensitivity of K on the NWPU VHR-10 dataset under a common development protocol.

K	Bbox (%)			Mask (%)		
	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
32	69.38	91.69	79.03	64.94	92.91	70.01
64	70.35	92.23	81.65	66.65	94.15	71.97
128	70.13	92.42	81.32	65.97	93.71	72.22

3.2. Comparative Analysis

PromptMiner-SAM2 is compared with 10 representative instance-segmentation methods. They are grouped into CNN-based methods (CNN) and global sequence modeling methods (GSM). CNN includes two-stage, one-stage, and cascade

designs, whereas GSM includes Transformer-style mask prediction and foundation-model prompting. The *Type* column in Table 2 uses these two abbreviations.

It can be concluded from Table 2 that our method obtains the highest mask AP on the WHU dataset and the NWPU VHR-10 dataset, leading by 0.1 and 2.2 points, respectively, over the strongest alternatives. Its WHU mask AP_{75} is also the best, while the bbox results remain close to the leading YOLO11s-seg baseline. On the NWPU VHR-10 dataset, it leads the bbox AP, bbox AP_{75} , mask AP, and mask AP_{50} comparisons; although mask AP_{75} is below the best value, it remains competitive with other GSM methods.

Figure 3 compares predictions on the WHU dataset and the NWPU VHR-10 dataset. In the WHU row, touching roofs remain separated; in the NWPU row, object silhouettes are more complete with fewer merges.

Table 3. Prompt-path ablation on the NWPU VHR-10 dataset; *P*, *B*, and *M* denote point, box, and dense-mask prompts, and UDPR denotes uncertainty-guided refinement.

Variant	Bbox (%)			Mask (%)		
	AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}
<i>P</i>	71.81	93.60	81.89	63.15	92.56	67.73
<i>B</i>	72.14	94.34	82.24	65.25	94.12	69.02
<i>P+B</i>	71.74	93.52	81.68	66.94	93.93	72.73
<i>P+B+M</i>	73.76	93.80	84.69	66.35	93.66	71.61
<i>P+B+M</i> + UDPR	73.43	93.96	84.16	68.28	94.53	74.61

3.3. Ablation Study

The effects of adding prompt types and decoder-tail refinement are summarized in Table 3. With only the box prompt *B*, mask AP is increased from 63.15% to 65.25%, and mask AP_{75} from 67.73% to 69.02%. With both discrete prompts (*P+B*), 66.94% mask AP and 72.73% mask AP_{75} are obtained, showing complementary cues.

With the dense-mask prompt, 66.35% mask AP and 71.61% mask AP_{75} are obtained, while bbox AP and bbox AP_{75} are 73.76% and 84.69%, respectively. For the complete *P+B+M+UDPR* configuration, 68.28% mask AP and 74.61% mask AP_{75} are obtained, the best mask results in this ablation.

4. CONCLUSION

This paper presents PromptMiner-SAM2, a framework that addresses ambiguous instance separation in overhead imagery by converting proposal-local shape cues into complementary prompts for SAM2 and refining uncertain mask locations. The resulting design improves remote-sensing instance segmentation, reaching 73.4% mask AP on the WHU dataset and 68.3% on the NWPU VHR-10 dataset. Future work can

explore richer prompt-mining schemes, including cues that adapt to object scale and local scene context.

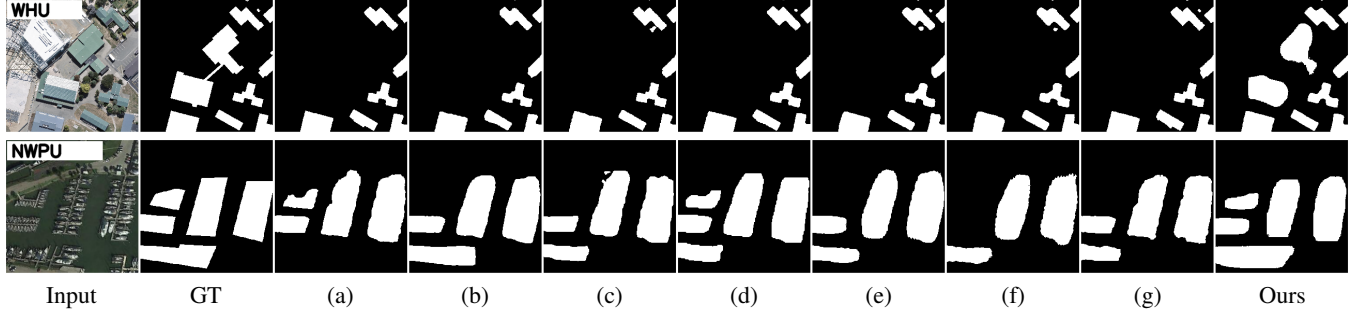


Fig. 3. Qualitative comparison on the WHU and NWPU VHR-10 datasets. From left to right, the columns show the input image, ground truth, predictions of (a) MaskDINO, (b) RSPrompter, (c) CondInst, (d) YOLO11, (e) HTC, (f) Mask R-CNN, (g) M2F, and PromptMiner-SAM2 (ours).

Table 2. Comparative analysis on the WHU dataset and the NWPU VHR-10 dataset. Each entry reports AP / AP₅₀ / AP₇₅ (%) for the Bbox and Mask columns.

Type	Method	WHU						NWPU VHR-10					
		Bbox (%)			Mask (%)			Bbox (%)			Mask (%)		
		AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
CNN	Mask R-CNN [3]	70.9	89.9	80.9	68.2	90.0	80.0	65.4	90.2	78.9	63.0	90.1	71.2
	MS R-CNN [4]	68.8	89.4	80.0	67.7	89.6	79.5	67.8	94.3	83.8	64.7	93.6	80.8
	HTC [5]	72.7	90.6	82.5	69.2	90.7	81.6	58.9	86.9	67.7	51.3	84.6	53.3
	CondInst [6]	74.2	90.0	82.4	69.7	90.2	80.9	65.0	90.1	79.6	62.8	90.7	68.1
	YOLO11s-seg [23]	76.7	91.2	85.4	66.8	91.3	79.1	71.0	94.4	79.5	63.9	93.9	67.3
GSM	Mask2Former [7]	61.4	82.7	70.7	64.5	84.1	75.6	60.3	77.9	66.8	61.5	84.9	67.8
	MaskDINO [8]	73.3	90.8	82.7	73.3	90.3	82.5	65.7	88.9	74.2	65.8	89.6	71.5
	RS4D-box [9]	64.5	87.6	74.7	64.1	87.8	75.2	57.0	82.5	65.4	54.3	80.1	57.6
	RSIISN [1]	74.1	90.0	84.4	72.1	90.1	83.9	66.7	91.5	78.4	62.9	89.7	67.1
	RSPrompter [13]	72.5	91.0	81.7	72.5	92.0	82.9	70.3	93.6	81.0	66.1	92.7	70.6
Ours	PromptMiner-SAM2	76.4	91.5	85.4	73.4	91.4	85.5	73.4	94.0	84.2	68.3	94.5	74.6

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